

Project Initialization and Planning Phase

Date	2 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service. Below, the problem statements for our credit approval solution are defined across key institutional and consumer roles to address operational bottlenecks, compliance risks, and customer experience barriers.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 <i>(Analyst Console)</i>	A junior credit risk analyst under pressure to process backlogs of credit card applications at a commercial retail bank.	Perform quick, accurate creditworthiness evaluations on applicant financial profiles to make credit decisions within a strict 24-hour service level agreement (SLA).	Manual verification is tedious, relies on static guidelines, and is highly prone to human classification mistakes and subjective bias.	The daily incoming application volume is extremely high, and checking multi-variable applicant details manually is very slow and inefficient.	Overwhelmed by growing backlogs, anxious about accidentally approving high-risk defaults, and stressed by SLA target compliance reports.
PS-2 <i>(Compliance Audit)</i>	A senior bank compliance auditor responsible for portfolio health and risk management standards.	Audit applicant portfolios to identify and instantly reject high-risk delinquency records to maintain low default rates.	Applicants with past-due payment histories slip through automated approvals, introducing bad debt risk into the active portfolio.	Credit registry records utilize complex multi-class payment status codes that are not integrated as hard real-time rule overrides.	Concerned about auditing failures, regulatory non-compliance fines, and increases in the bank's non-performing loan ratio.

PS-3 <i>(Eligibility Check)</i>	A prospective credit card applicant seeking a credit limit to manage daily expenses.	Check pre-qualification approval odds and credit limit eligibility through an online bank portal.	I am hesitant to check my eligibility out of fear of hurting my credit score through hard pull inquiry rejections.	Standard bank qualification checks submit hard inquiries to credit bureaus, which instantly damages credit score histories.	Reluctant to apply, anxious about receiving rejections, and insecure about credit score health.
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Key Objectives of the Credit Card Approval Prediction System:

To solve the problem statements defined above, the proposed system implements an intelligent machine learning decision engine backed by a premium responsive user dashboard. The core objectives include:

- 1. Machine Learning Multi-Model Comparison:** The system trains and compares four classification algorithms (Logistic Regression, Decision Tree, Random Forest, and XGBoost) on historical credit card applicants' demographics and credit histories. To maximize predictive accuracy, the model with the highest test accuracy is automatically selected as the active model to serve real-time scoring.
- 2. Seamless Real-Time Inference (Analyst Console):** Explains prediction decisions instantly. When an applicant is evaluated, the system provides a likelihood percentage and visualizes positive and negative contributors, giving the analyst immediate, explainable insights to make rapid decisions.
- 3. Delinquency Compliance Rules:** Implements hard-coded compliance overrides. The pipeline converts multi-class payment history status codes into a binary High Risk flag. Applicants with active delinquency records (status codes 1-5 representing past-due payments) are blocked automatically, protecting the bank from high-default profiles.
- 4. Soft-Check Customer Pre-Qualification:** Integrates a multi-step eligibility check wizard. This allows consumers to test their approval likelihood and receive personalized financial tips without triggering hard inquiries at credit bureaus.